



Catalog Number 6120418

3700sc Digital Conductivity Sensor

USER MANUAL

October 2006, Edition 4

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Section 1 Specifications

Specifications are subject to change without notice.

Table 1 Inductive Conductivity Probe Specifications

Components	Corrosion-resistant materials, fully-immersible probe with 6 m (20 ft) cable
Conductivity Measuring Range	0.0 to 200.0; 0 to 2,000,000 microSiemens/cm
Measuring Range (Temperature)	–10 to 200.0 °C (–14 to 392 °F) limited by sensor body material
Probe Operating Temperature	–10 to 200 °C (–14 to 392 °F); limited only by sensor body material and mounting hardware
Pressure/Temperature Limits (in pure water only)	Polypropylene: 6.9 bar at 100 °C (100 psi at 212 °F) PVDF: 6.9 bar at 120 °C (100 psi at 248 °F) PEEK ¹ : 13.8 bar at 200 °C (200 psi at 392 °F) PFA Teflon ² : 13.8 bar at 200 °C (200 psi at 392 °F)
Maximum Flow Rate	3 m (10 ft) per second
Temperature Compensator	PT 1000 RTD
Sensor Cable	Polypropylene and PVDF sensors: 5-conductor (plus two isolated shields) with XLPE (cross-linked polyethylene) jacket; rated to 150 °C (302 °F); 20 ft (6 m) long PEEK ¹ and PFA Teflon ² sensors: 5-conductor (plus two isolated shields) with Teflon ² -coated jacket; rated to 200 °C (392 °F); 6 m (20 ft) long.
Probe Dimensions	See Figure 1 and Figure 2 on page 8 .

¹ PEEK[®] is a registered trademark of ICI Americas, Inc.

² Teflon[®] is a registered trademark of Dupont Co.

Table 2 Digital Gateway Specifications

Weight	145 g (5 oz)
Dimensions	17.5 x 3.4 cm (7 x 1 ³ / ₈ in.)
Operating Temperature	–20 to 60 °C (–4 to 140 °F)

2.1 Safety Information

Please read this entire manual before unpacking, setting up, or operating this equipment. Pay attention to all danger and caution statements. Failure to do so could result in serious injury to the operator or damage to the equipment.

To ensure that the protection provided by this equipment is not impaired, do not use or install this equipment in any manner other than that specified in this manual.

This product is acceptable for use in a Hazardous Location when used with an sc100 Controller and installed per Control Drawing 58600-78 as described in the sc100 Controller Manual, Cat. No. 5860018.

2.1.1 Use of Hazard Information

DANGER

Indicates a potentially or imminently hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION

Indicates a potentially hazardous situation that may result in minor or moderate injury.

Important Note: *Information that requires special emphasis.*

Note: *Information that supplements points in the main text.*

2.1.2 Precautionary Labels

Read all labels and tags attached to the instrument. Personal injury or damage to the instrument could occur if not observed

	This symbol, if noted on the instrument, references the instruction manual for operation and/or safety information.
	This symbol, when noted on a product enclosure or barrier, indicates that a risk of electrical shock and/or electrocution exists.
	This symbol, if noted on the product, indicates the need for protective eye wear.
	This symbol, when noted on the product, identifies the location of the connection for Protective Earth (ground).
	This symbol, when noted on the product, identifies the location of a fuse or current limiting device.

2.2 General Sensor Information

Optional equipment, such as mounting hardware for the probe, is supplied with instructions for all user installation tasks.

Figure 1 Sanitary-style Sensor

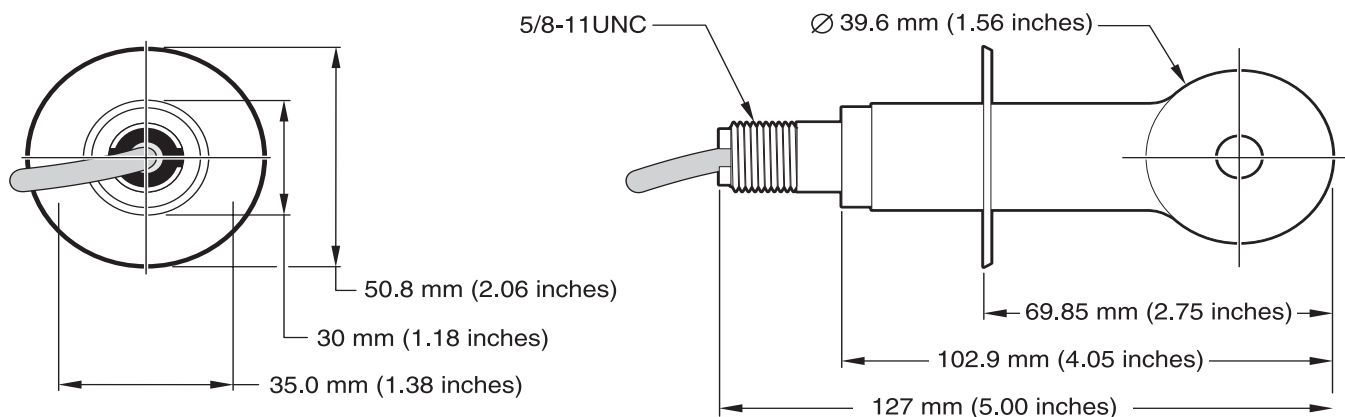
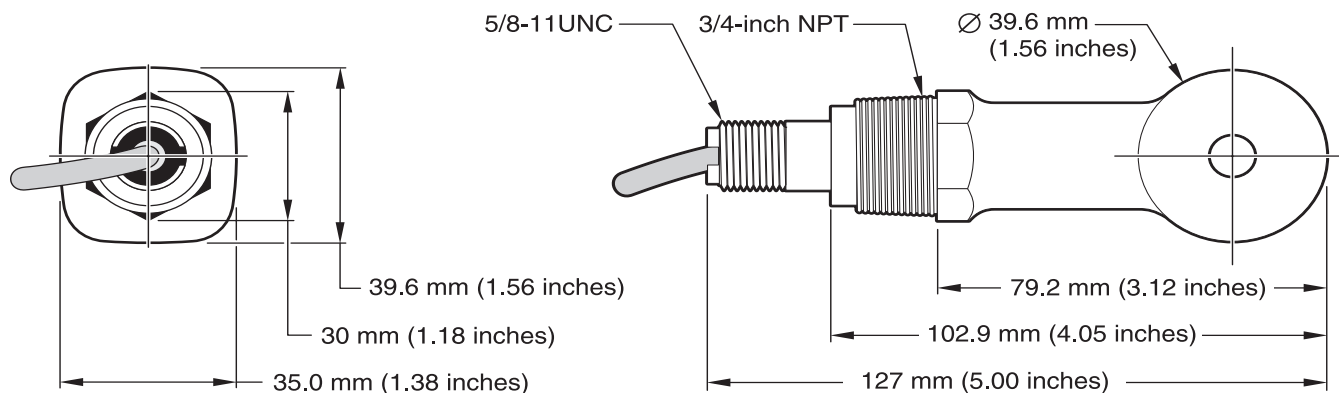


Figure 2 Convertible-style Sensor



2.3 The Digital Gateway

The Digital Gateway was developed to provide a means to use existing analog sensors with the new digital controllers. The gateway contains all the necessary software to interface with the controller and output a digital signal. Extension cables are required for connection from the digital gateway to the digital controller. See [Replacement Parts and Accessories on page 33](#).

2.4 Theory of Operation

Inductive conductivity measurements are made by passing an AC current through a toroidal drive coil which induces a current in the electrolyte solution. This induced solution current produces a current in a second toroidal coil. The amount of current induced in the second coil is proportional to the solution conductivity.

The major advantage of toroidal conductivity is that the coils are not in contact with the solution. Because the coils are isolated from the solution, oils and other contaminants in moderate amounts do not interfere with the measurement. In addition, the coil housing can be chosen to withstand corrosive environments which would quickly corrode traditional sensors with metal electrodes.

DANGER

Only qualified personnel should conduct the tasks described in this section of the manual.

DANGER

Seul un technicien qualifié peut effectuer les tâches d'installation décrites dans cette section du manuel.

The 3700sc Digital Conductivity Sensor can be used with either an sc100 or sc1000 controller. Refer to [section Figure 4 on page 10](#) for sc100 installation instructions and [section 3.3 on page 14](#) for sc1000 installation instructions.

The sc sensor should be wired to the digital gateway before connecting it to the sc100 or sc1000 Controller. The digital gateway is designed to provide a digital interface to the appropriate digital controller. Refer to [section 3.1](#) for more information.

3.1 Wiring the sc Sensor to the Digital Gateway

DANGER

The sc100 and certain versions of the sensor are suitable for use in Class 1, Division 2, Groups A, B, C, D Hazardous Locations . See Control Drawing 58600-78 in the sc100 Controller Manual, Cat. No. 58600-18 for acceptable sensor versions and installation requirements.

DANGER

Le sc100 et certaines versions du capteur peuvent être utilisés dans des endroits dangereux de la Classe 1, Division 2, Groupes A, B, C, D. Reportez-vous au schéma de contrôle 58600-78 du Manuel du contrôleur sc100, Réf. 58600-18 pour connaître les versions des capteurs admises et les conditions d'installation.

DANGER

Explosion hazard. Do not connect or disconnect equipment unless power has been switched off or the area is known to be non-hazardous.

DANGER

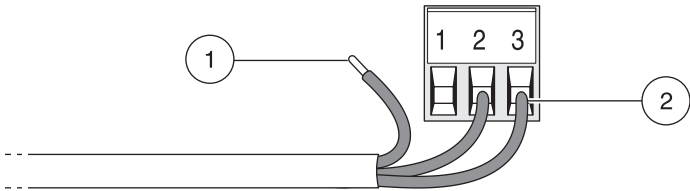
Risque d'explosion. Couper le courant ou s'assurer que l'emplacement est désigné non dangereux avant de remplacer le aucun composant.

1. Route the cable from the sensor through the strain relief in the digital gateway then properly terminate the wire ends (see [Figure 3](#)).

Note: Do not tighten the strain relief until the digital gateway is wired and the two halves are threaded securely together.

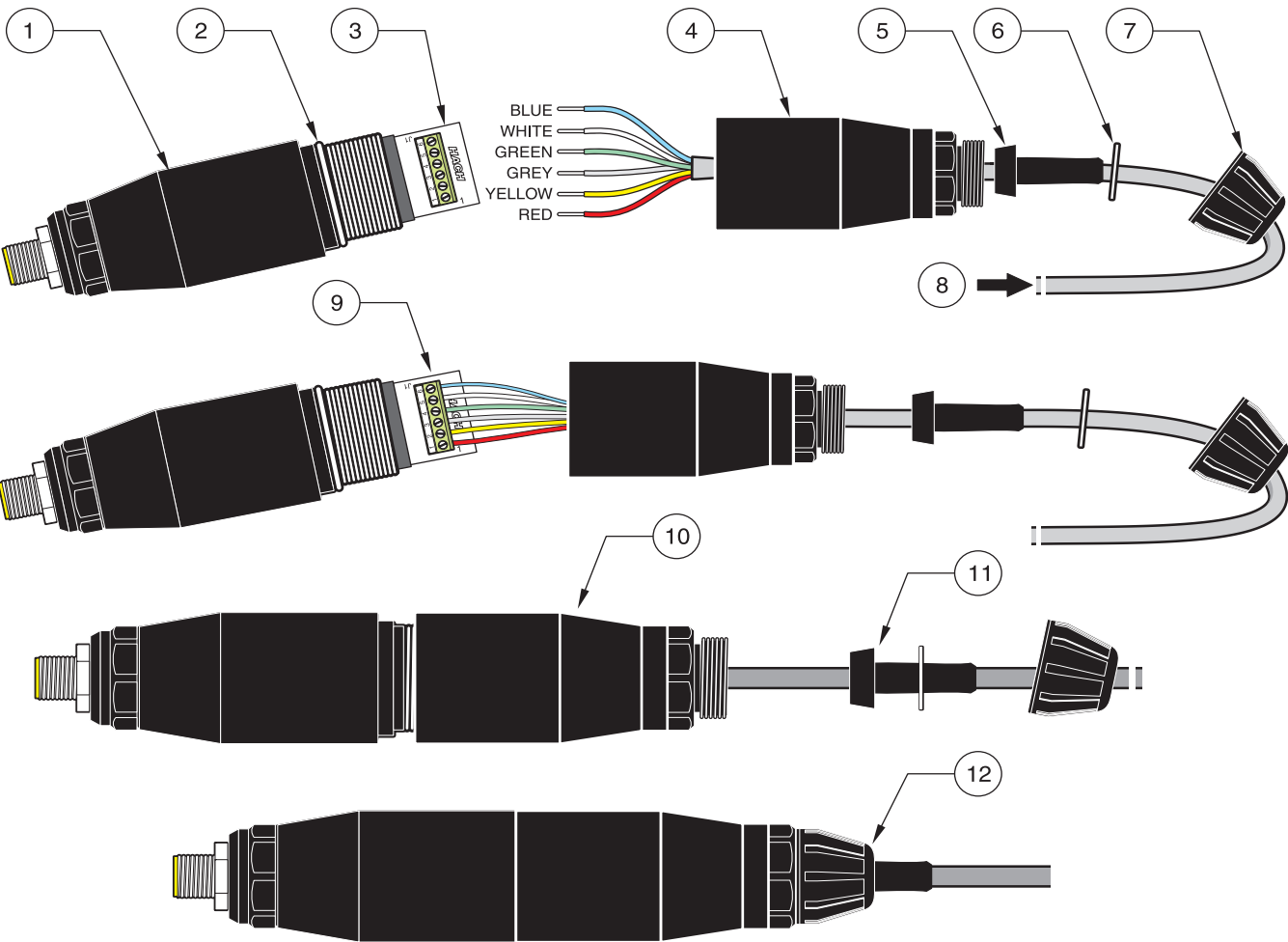
2. Insert the wires as shown in [Table 3](#) and [Figure 4](#).
3. Make sure the O-ring is properly installed between the two halves of the digital gateway and thread the two halves together. Hand tighten.
4. Tighten the strain relief to secure the sensor cable.
5. Connect the digital gateway to the controller.
 - sc100 Non-hazardous Location—[section 3.2.1 on page 11](#)
 - sc100 Hazardous Location—[section 3.2.2 on page 13](#)
 - sc1000 Connection Instructions—Refer to [section 3.3 on page 14](#).

Figure 3 Proper Wire Preparation and Insertion



- | | |
|--------------------------------|---|
| 1. Strip ¼-inch of insulation. | 2. Seat insulation against connector with no bare wire exposed. |
|--------------------------------|---|

Figure 4 Wiring and Assembling the Digital Gateway



1. Digital gateway front	7. Nut, strain relief
2. O-ring	8. From sensor
3. Sensor wire connector	9. Insert wires into the connector according to Table 3 . Use the included 2-mm screwdriver (Cat. No. 6134300) to secure connections.
4. Digital gateway back	10. Screw back of digital gateway onto front
5. Cable bushing	11. Push cable bushing and anti-rotation washer into back.
6. Anti-rotation washer	12. Fasten cord grip securely. Assembly is complete.

Table 3 Wiring the Digital Gateway

Sensor (wire color)	Sensor Signal	Digital Gateway Sensor Wire connector
Red	Temp +	1
Yellow	Temp –	2
Shield (grey)	Shield	3
Green	Sense	4
White	Drive +	5
Blue	Drive –	6

3.2 Connecting/Wiring the Digital Gateway to the sc100 Controller

DANGER

The sc100 and certain versions of the sensor are suitable for use in Class 1, Division 2, Groups A, B, C, D Hazardous Locations . See Control Drawing 58600-78 in the sc100 Controller Manual, Cat. No. 58600-18 for acceptable sensor versions and installation requirements.

DANGER

Le sc100 et certaines versions du capteur peuvent être utilisés dans des endroits dangereux de la Classe 1, Division 2, Groupes A, B, C, D. Reportez-vous au schéma de contrôle 58600-78 du Manuel du contrôleur sc100, Réf. 58600-18 pour connaître les versions des capteurs admises et les conditions d'installation.

The digital gateway should be wired to the sensor before connecting to the controller.

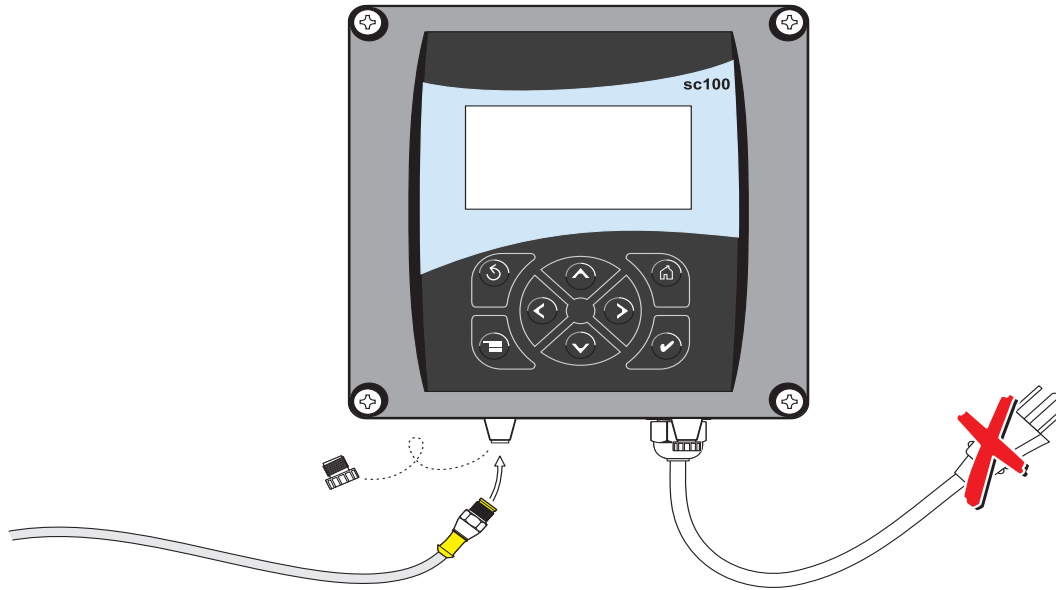
3.2.1 Connecting the Digital Gateway in a Non-hazardous Location

3.2.1.1 Attaching the Digital Gateway with a Quick-connect Fitting

Important Note: The standard quick-connect fitting is NOT suitable for Class 1, Division 2 Hazardous Location installations without the connector lock installed, see [section 3.2.1 on page 11](#) for more information.

The sensor has a keyed quick-connect fitting for easy attachment to the controller, see [Figure 5](#). Retain the connector cap to seal the connector opening when the sensor is removed. Extension cables may be purchased to extend the sensor cable length. If the total cable length exceeds 100 m (300 ft), a termination box must be installed. See [Replacement Parts and Accessories on page 33](#).

Figure 5 Attaching the Digital Gateway using the Quick-connect Fittings



3.2.1.2 Hard-wiring the Digital Gateway to the sc100

Important Note: Hard-wiring the digital gateway to the sc100 is not an approved method for Class 1, Division 2 Hazardous Locations.

1. Disconnect power to the controller if powered.
2. Open the controller cover.
3. Disconnect and remove the existing wires between the quick-connect and terminal strip J5, see [Figure 6 on page 13](#).
4. Remove the quick-connect fitting and wires and install the threaded plug on the opening to maintain the environmental rating.
5. Cut the connector from the sensor cable.
6. Strip the insulation on the cable back 1-inch. Strip ¼-inch of each individual wire end.
7. Pass the cable through conduit and a conduit hub or a strain relief fitting (Cat. No. 16664) and an available access hole in the controller enclosure. Tighten the fitting.

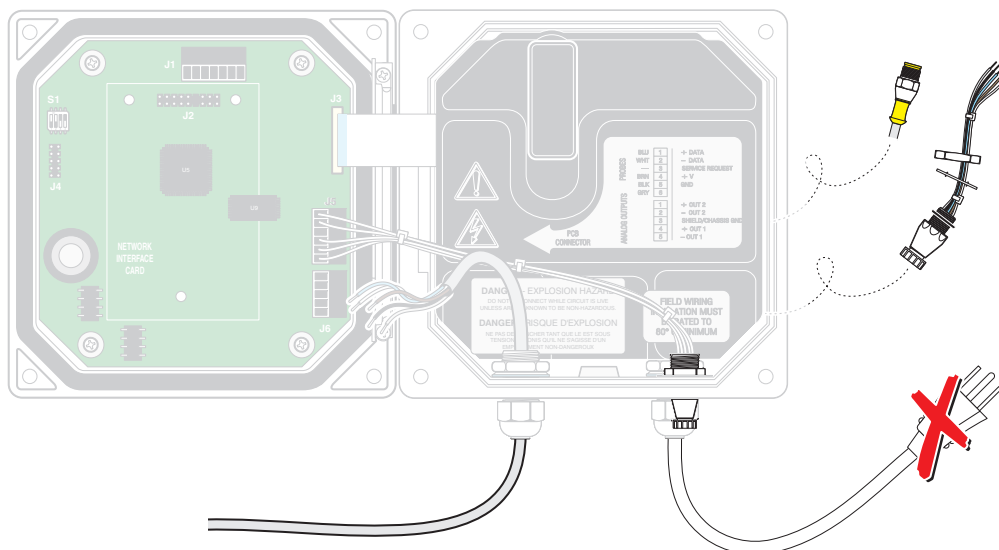
Note: Use of strain relief fitting other than Cat. No. 16664 may result in a hazard. Use only the recommended strain relief fitting.

8. Reinstall the plug on the sensor access opening to maintain the environmental rating.
9. Wire as shown in [Table 4](#) and [Figure 6](#).
10. Close and secure the cover.

Table 4 Wiring the Sensor at Terminal Block J5

Terminal Number	Terminal Designation	Wire Color
1	Data (+)	Blue
2	Data (-)	White
3	Service Request	No Connection
4	+12 V dc	Brown
5	Circuit Common	Black
6	Shield	Shield (grey wire in existing quick disconnect fitting)

Figure 6 Hard-wiring the Digital Gateway



3.2.2 Connect the Digital Gateway Sensor to a sc100 Controller in a Hazardous Location

DANGER

The sc100 and certain versions of the sensor are suitable for use in Class 1, Division 2, Groups A, B, C, D Hazardous Locations. See Control Drawing 58600-78 in the sc100 Controller Manual, Cat. No. 58600-18 for acceptable sensor versions and installation requirements.

DANGER

Le sc100 et certaines versions du capteur peuvent être utilisés dans des endroits dangereux de la Classe 1, Division 2, Groupes A, B, C, D. Reportez-vous au schéma de contrôle 58600-78 du Manuel du contrôleur sc100, Réf. 58600-18 pour connaître les versions des capteurs admises et les conditions d'installation.

DANGER

Explosion hazard. Do not connect or disconnect equipment unless power has been switched off or the area is known to be non-hazardous.

DANGER

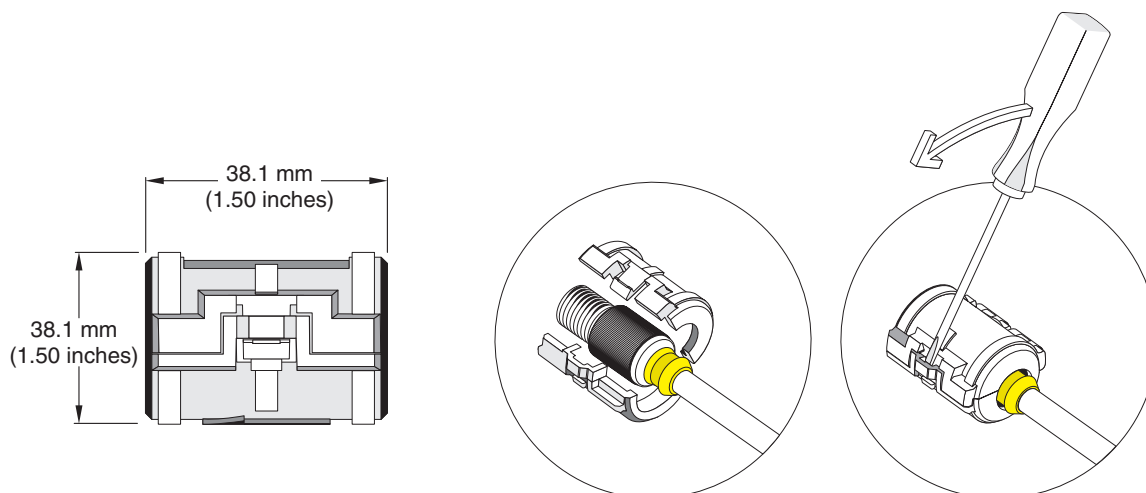
Risque d'explosion. Couper le courant ou s'assurer que l'emplacement est désigné non dangereux avant de remplacer le aucun composant.

3.2.2.1 Attaching the Digital Gateway with Quick-Connect Fitting in a Hazardous Location

The digital gateway is supplied with a keyed quick-connect fitting for easy attachment to the controller, see [Figure 5 on page 12](#). For Hazardous Locations, a connector safety lock (Cat. No. 6139900) must be installed. Retain the connector cap to seal the connector opening in case the sensor must be removed.

1. Remove the connector cap from the sc100 controller. Retain the connector cap to seal the connector opening in case the sensor must be removed.
2. Connect the digital gateway connector to the sensor plug on the sc100.
3. Install the connector safety lock ([Figure 7](#)). Align the lock over the connector and squeeze the two halves together to lock. To remove the connector safety lock, insert a small flat-blade screwdriver into the locking groove. Pivot the screwdriver away from the groove and separate the two halves ([Figure 7](#)).

Figure 7 **Installing the Connector Safety Lock**



3.3 Connecting the Sensor to the sc1000

3.3.1 Connecting the Sensor using the Quick-connect Fittings

1. Unscrew the protective cover on the controller socket. Retain the protective cover for the sensor. When the sensor is removed, re-fit the protective cover.
2. Push the connector into the socket. During this process pay attention to the lugs on the connector.
3. Hand-tighten the union nut.

Note: Do not use the middle connection for the sensors, which is reserved for the display module.

3.4 Mounting the Digital Gateway

The digital gateway is supplied with a mounting clip for mounting to a wall or other flat surface. Digital gateway dimensions are shown in [Figure 8](#). Use an appropriate fastener to secure it to the wall. After the sensor is wired to the digital gateway and the two halves are threaded together, place the mounting clip over the center of the digital gateway and squeeze the clip together to secure. See [Figure 9](#).

Figure 8 Digital Gateway Dimensions

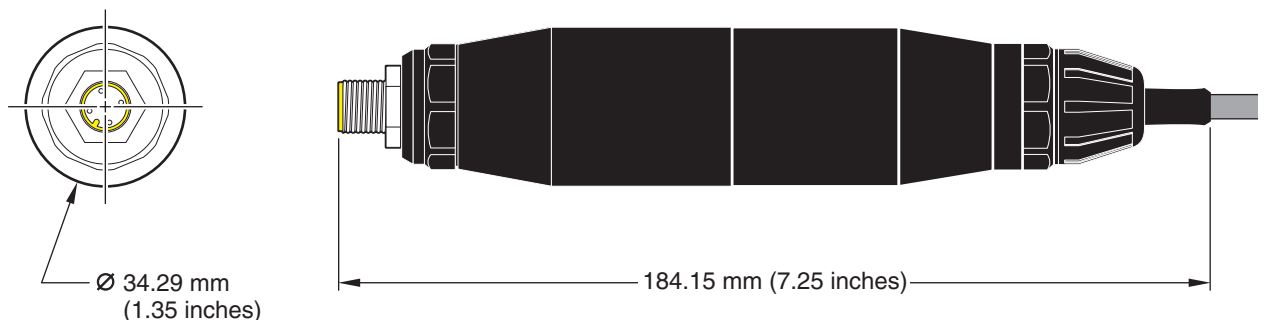
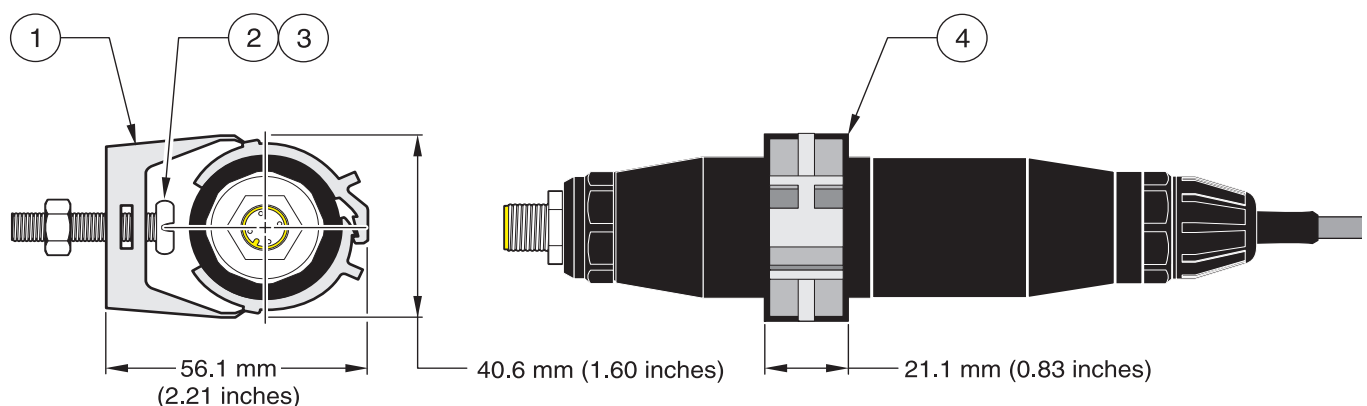


Figure 9 Mounting the Digital Gateway



1. Mounting clip	3. Hex nut, 1/4-28
2. Screw, pan head, 1/4-28 x 1.25-in.es	4. Mount clip, insert digital gateway, squeeze clip closed

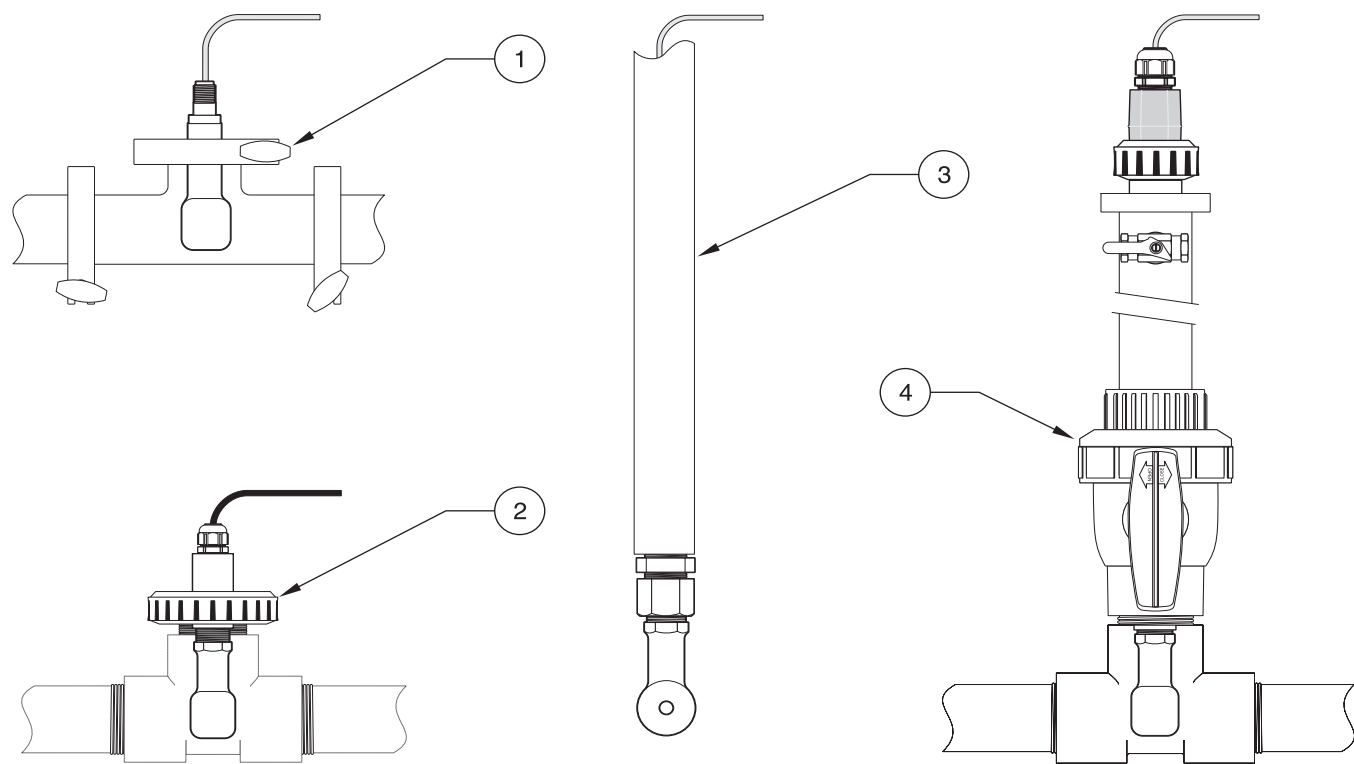
3.5 Installing the Sensor in the Sample Stream

Locate the sensor as close a possible to the instrument. The convertible style sensor may be immersion mounted by threading it onto the end of a pipe of an appropriate length. It may also be mounted in any standard 2-in. NPT pipe tee, weldolet, or pipe saddle using the special union mount adapter. In addition, the sensor may be installed in a ball valve in a pressurized or non-pressurized process pipe/vessel.

The sanitary-style sensor may be clamp-mounted for clean-in-place applications using a sanitary clamp-type tee or ferrule and a heavy-duty clamp.

Examples of common sensor installations are shown in [Figure 10](#) and dimension drawings are shown in [Figure 1](#) and [Figure 2 on page 8](#). Refer to the instructions supplied with the mounting hardware for installation specifics.

Figure 10 Sensor Installation Examples



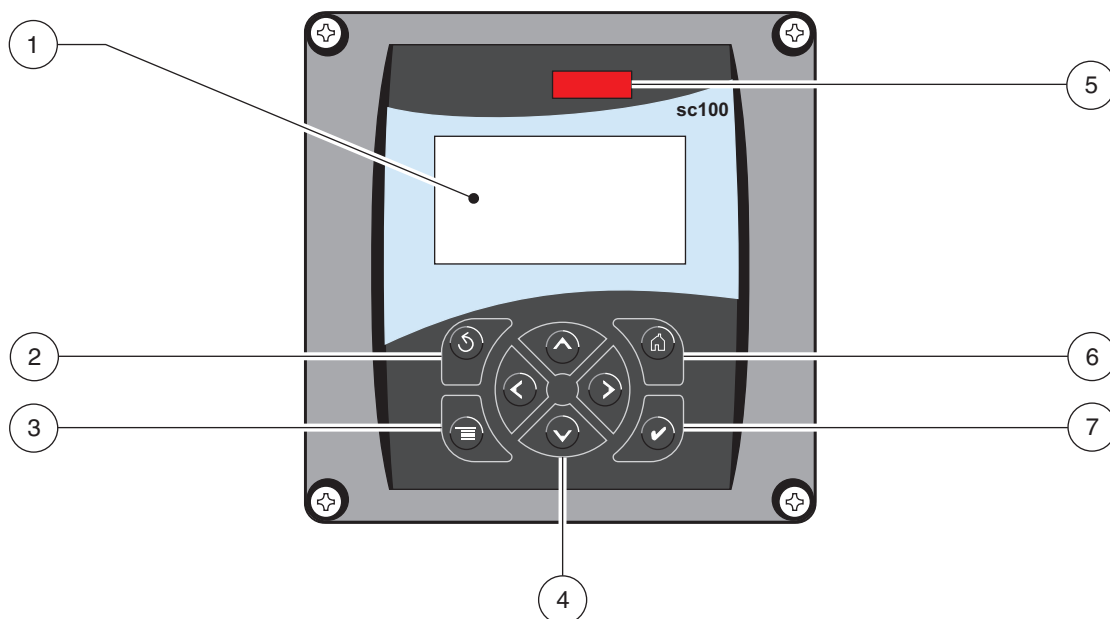
1. Sanitary-style sensor sanitary clamp mounting	3. Convertible-style sensor immersion mounting
2. Convertible-style sensor union mounting	4. Convertible-style sensor ball valve mounting

Section 4 User Interface and Navigation

4.1 Using the sc100 Controller

The front of the controller is shown in [Figure 11](#). The keypad consists of the eight keys described in [Table 5](#).

Figure 11 Front of the Controller



1. Instrument display	5. IrDA window
2. BACK key	6. HOME key
3. MENU key	7. ENTER key
4. RIGHT, LEFT, UP, and DOWN keys	

Table 5 Controller Key Functions/Features

Number	Key	Function
2		Moves back one level in the menu structure.
3		Moves to the main menu from other menus. This key is not active in menus where a selection or other input must be made.
4		Navigates through the menus, changes settings, and increments and decrements digits.
5		Moves to the Main Measurement screen from any other screen. This key is not active in menus where a selection or other input must be made.
6		Accepts an input value, updates, or accepts displayed menu options.

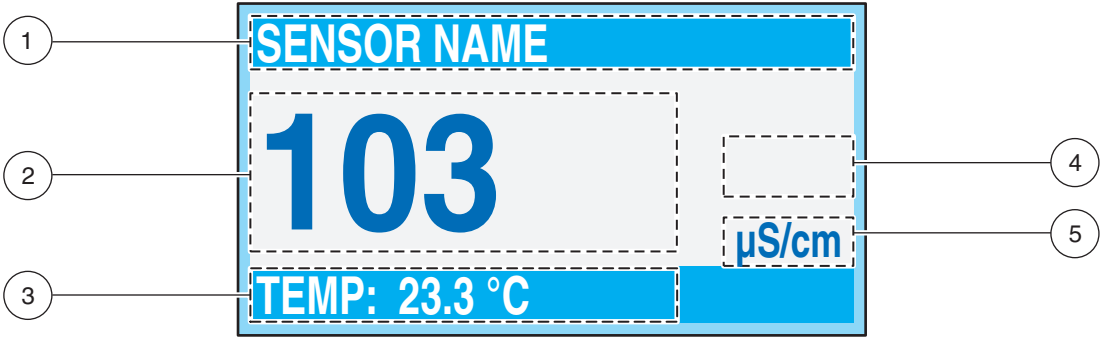
4.1.1 Controller Display Features

When a sensor is connected and the controller is in measurement mode, the controller display will show the current conductivity reading plus the sample temperature.

The display will flash on startup, when a sensor error has occurred, when the hold outputs function has been activated, and when a sensor is being calibrated.

An active system warning will cause the warning icon (a triangle with an exclamation point inside) to be displayed on the right side of the display.

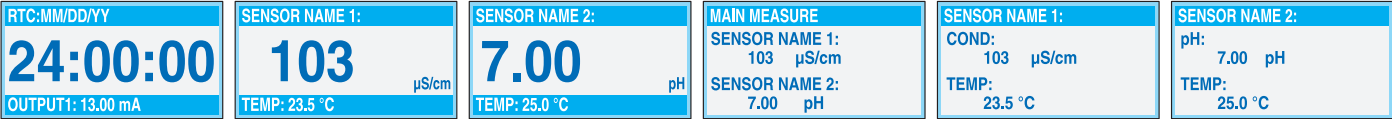
Figure 12 Display



1. Status bar. Indicates the sensor name and status of relays. The relay letter is displayed when the relay is energized.	3. Secondary measurement
2. Main measurement	4. Warning icon area
	5. Measurement units (μS, mS, S, mohm, TDS)

4.1.2 Important Key Presses

- Press the **HOME** key then the **RIGHT** or **LEFT** key to display two readings when two sensors are connected. Continue to press the **RIGHT** or **LEFT** key to toggle through the available display options as shown below.



- Press the **UP** and **DOWN** keys to toggle the status bar at the bottom of the measurement display to display the secondary measurement (temperature) and output information.



- When in Menu mode, an arrow may appear on the right side of the display to indicate that more menus are available. Press the **UP** or **DOWN** key (corresponding to the arrow direction) to display additional menus.



4.2 Using the sc1000 Controller

The sc1000 is a touch screen application. Use your finger to touch keys and menu commands. In normal operation the touch screen displays the measured values for the sensors selected.

4.2.1 Display Features

4.2.1.1 Using the Pop-up Toolbar

The pop-up toolbar provides access to the controller and sensor settings. The toolbar is normally hidden from view. To view the toolbar, touch the bottom-left of the screen.

Figure 13 Pop-up Toolbar Functions



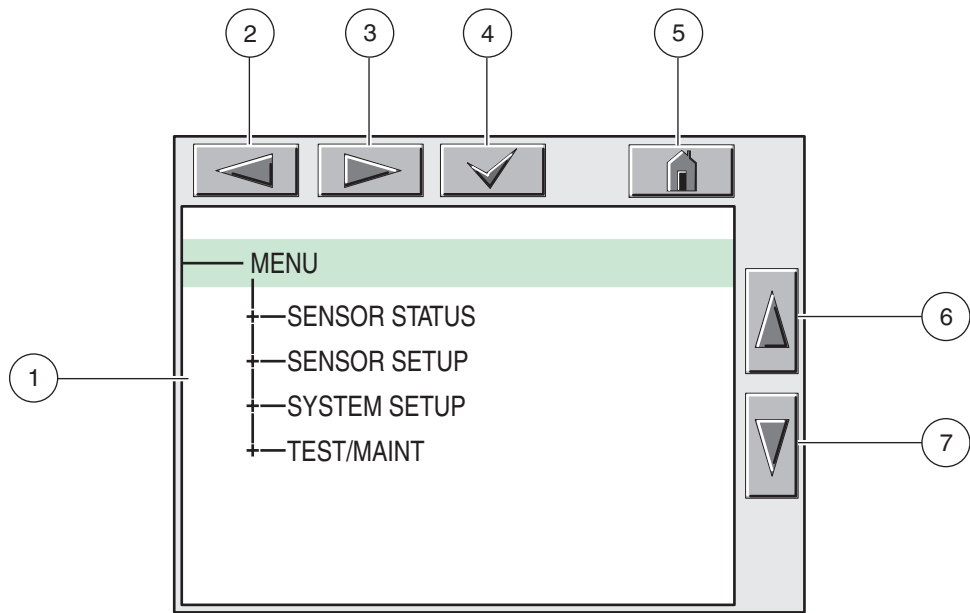
	MAIN MENU —displays the Main Menu Structure
	UP Arrow —scrolls up to the previous displayed value.
	Displays one value.
	Displays two values at the same time.
	Displays four values at the same time.
	LIST —displays the list of connected devices and sensors.
	DOWN Arrow —scrolls down to the next displayed value.

4.2.1.2 Using the Menu Windows

If the Menu button (from the pop-up toolbar) is selected, the Main Menu screen is opened. The Main Menu screen allows the user to view the sensor status, configure the sensor setup, system setup, and perform diagnostics.

The menu structure may vary depending on the configuration of the system.

Figure 14 Main Menu



1.	Display Area
2.	BACK
3.	FORWARD
4.	ENTER —confirms the entry or selection.
5.	HOME —changes to the display of measured values. The pop-up toolbar cannot open from the menu window. To view the Main Menu from this display, touch the Home button and then the bottom of the screen.
6.	UP —scrolls up
7.	DOWN —scrolls down

4.2.1.3 Navigating the Menu Windows

To view a menu item, touch the menu item or use the **UP** and **DOWN** keys to highlight the item. The menu item remains highlighted for approximately 4 seconds after it is selected. To view the highlighted command, select the area to the left of the menu item or select the **ENTER** button.

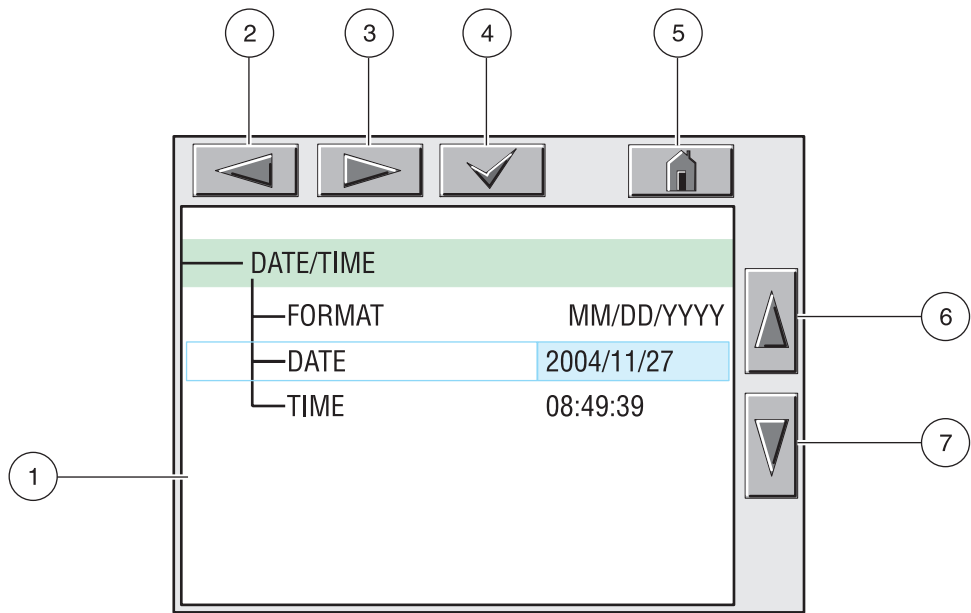
A “+” next to a menu command indicates there is a submenu. Touch the “+” to view the submenu. An “i” next to a menu command indicates it is information only.

If a menu item is editable, highlight the item and touch the far-left part of the menu item until it is highlighted and press **ENTER** or double-tap the highlighted item. A keypad will be displayed to change an entry (Figure 16 on page 21) or a list box will be displayed (Figure 17 on page 22).

Messages are displayed in the message window (Figure 18 on page 22).

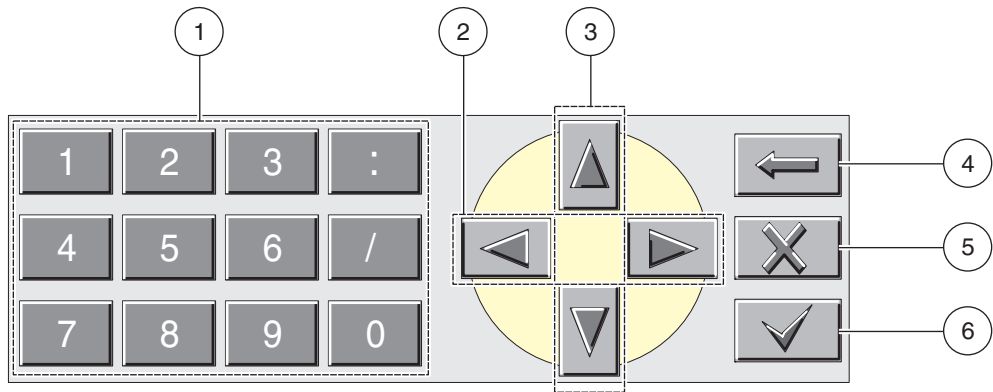
If an entry is incorrect, repeat the entry with the correct values. If the entry is outside the working range, a correction to the entry is made automatically.

Figure 15 Changing a Menu Item



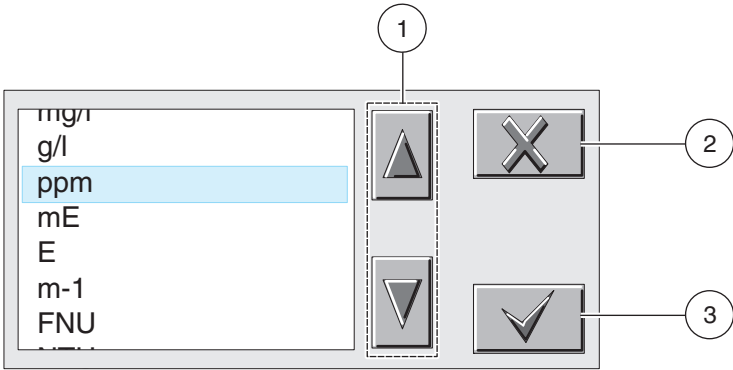
1. Display Area	5. HOME—changes to the display of measured values.
2. BACK	6. UP—scrolls up
3. FORWARD	7. DOWN—scrolls down
4. ENTER—confirms the entry or selection.	

Figure 16 Keypad



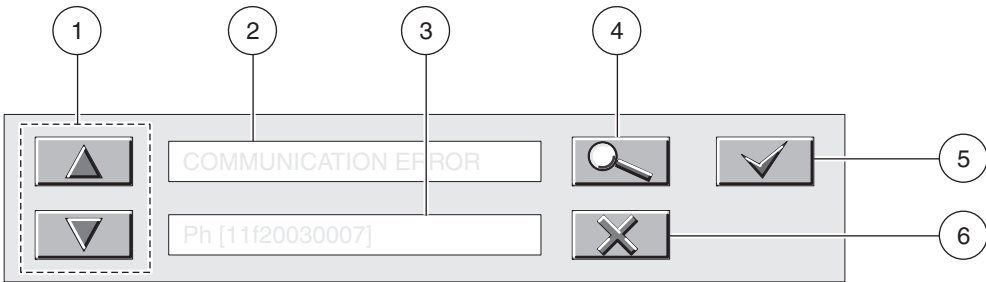
1. Enters numbers or the character as shown on the button.
2. Moves the cursor one position to the left or to the right.
3. Increase/Decrease a number or letter at the cursor position. Keep the button pressed to change the numbers/characters continuously.
4. Deletes the character to the left of the cursor.
5. CANCEL—cancels the entry.
6. ENTER—confirms the entry or selection.

Figure 17 List Box



- | |
|--|
| 1. Scrolls up or down |
| 2. CANCEL —cancels and entry. |
| 3. ENTER —confirms a selection. |

Figure 18 Message window



- | |
|---|
| 1. Scrolls up or down. |
| 2. Displays the messages or warnings. |
| 3. Displays details on the selected entry. |
| 4. This button changes back to the previous display. |
| 5. ENTER —confirms an entry. |
| 6. CANCEL —cancels an entry. |

Section 5 Operation

5.1 Using the sc Controller

Before using the sensor in combination with an sc controller make yourself familiar with the operating mode of the controller. Refer to the controller user manual and learn how to use and navigate the menu functions.

5.2 Sensor Setup

When a sensor is initially installed, the serial number of the sensor will be displayed as the sensor name. To change the sensor name refer to the following instructions:

1. Select the Main Menu.
2. From the Main Menu, select SENSOR SETUP and confirm.
3. Select the appropriate sensor if more than one sensor is attached and confirm.
4. Select CONFIGURE and confirm.
5. Select EDIT NAME and edit the name. Confirm or cancel to return to the Sensor Setup menu.

5.3 Sensor Data Logging

The sc controller provides one data log and one event log for each sensor. The data log stores the measurement data at selected intervals. The event log stores a variety of events that occur on the devices such as configuration changes, alarms, warning conditions, etc. The data log and the event log can be read out in a CSV format. For downloading the logs please refer to the controller user manual.

5.4 Sensor Status Menu

SELECT SENSOR (if more than one sensor is attached)

STATUS	
ERROR LIST	See section 7.1 on page 31 .
WARNING LIST	See section 7.2 on page 31 .

5.5 Sensor Setup Menu

SELECT SENSOR (if more than one sensor is attached)

CALIBRATE	
ZERO	Perform a zero cal to remove sensor offset.
SAMPLE CAL	Perform a single point calibration.
COND CAL	After setting the reference temperature and slope, conduct a single-point calibration of the conductivity sensor.
TEMP ADJUST	Adjust the displayed temperature by up to ± 15 °C.
DEFAULT SETUP	Resets all user-editable options to their factory-defaults.

5.5 Sensor Setup Menu (continued)

CONFIGURE																																			
EDIT NAME	Enter a 10-digit name in any combination of symbols and alpha or numeric characters.																																		
SELECT MEASURE	Choose from Conductivity, Concentration, TDS, or Salinity. If Concentration is selected, the option to configure the concentration settings is offered. Press the down arrow to Config Conc. Two concentration types are offered: Built-in and User Defined. The Built-in menu offers an array of chemicals and concentrations to choose from: Built-in concentration tables <table><thead><tr><th>Solution</th><th>Concentration</th><th>°C Range</th></tr></thead><tbody><tr><td>NaOH</td><td>0–16%</td><td>0–100 °C</td></tr><tr><td>CaCl₂</td><td>0–22%</td><td>15–55 °C</td></tr><tr><td>HNO₃</td><td>0–28%</td><td>0–50 °C</td></tr><tr><td>HNO₃</td><td>36–96%</td><td>0–50 °C</td></tr><tr><td>H₂SO₄</td><td>0–30%</td><td>0–115 °C</td></tr><tr><td>H₂SO₄</td><td>40–80%</td><td>0–115 °C</td></tr><tr><td>H₂SO₄</td><td>93–99%</td><td>0–115 °C</td></tr><tr><td>H₃PO₄</td><td>0–40%</td><td>0–75 °C</td></tr><tr><td>HCl</td><td>0–18%</td><td>0–65 °C</td></tr><tr><td>HCl</td><td>22–36%</td><td>0–65 °C</td></tr></tbody></table>		Solution	Concentration	°C Range	NaOH	0–16%	0–100 °C	CaCl ₂	0–22%	15–55 °C	HNO ₃	0–28%	0–50 °C	HNO ₃	36–96%	0–50 °C	H ₂ SO ₄	0–30%	0–115 °C	H ₂ SO ₄	40–80%	0–115 °C	H ₂ SO ₄	93–99%	0–115 °C	H ₃ PO ₄	0–40%	0–75 °C	HCl	0–18%	0–65 °C	HCl	22–36%	0–65 °C
Solution	Concentration	°C Range																																	
NaOH	0–16%	0–100 °C																																	
CaCl ₂	0–22%	15–55 °C																																	
HNO ₃	0–28%	0–50 °C																																	
HNO ₃	36–96%	0–50 °C																																	
H ₂ SO ₄	0–30%	0–115 °C																																	
H ₂ SO ₄	40–80%	0–115 °C																																	
H ₂ SO ₄	93–99%	0–115 °C																																	
H ₃ PO ₄	0–40%	0–75 °C																																	
HCl	0–18%	0–65 °C																																	
HCl	22–36%	0–65 °C																																	
COND UNITS	Choose from the displayed units (dependent on the parameter selected in the Select Measure menu).																																		
TEMP UNITS	Select Celsius or Fahrenheit.																																		
T-COMPENSATION	The factory default for temperature compensation is linear with a 2.00% per °C slope and a 25 °C reference temperature. The default settings are appropriate for most aqueous solutions. The menu options are different depending on the measurement that is selected in the Configure>Select Measure menu. The options are as follows: NONE: No temperature compensation. LINEAR: Recommended for most applications. Press ENTER to change the slope or reference temperature. NATURAL WATER: Contact Technical Consulting Services for application specific information and assistance. TEMP TABLE: Use to configure a temperature compensation table by entering up to 10 x-axis values and 10 y-axis values. Contact Technical Consulting Services for additional information and assistance.																																		
LOG SETUP	Choose from Sensor Interval or Temp Interval. If the interval is enabled, choose from the displayed options to specify the frequency to log the sensor or temperature reading. Default is Disabled.																																		
AC FREQUENCY	Select 50 Hertz or 60 Hertz to specify the frequency to be excluded.																																		
FILTER	Average the measurement over time by entering a number between 0–60. Default is 0 seconds.																																		
TEMP ELEMENT	Select the temperature element type (100PT, 1000PT (default), or manual) then choose Select Factor to enter the specific “T” Factor supplied with the sensor.																																		
CAL DAYS	Number of days since the last calibration. Default notification at 60 days.																																		
SENSOR DAYS	Number of days the sensor has been in operation. Default notification at 365 days.																																		
DEFAULT SETUP	Reset the configure settings to the factory defaults.																																		
DIAG/TEST																																			
PROBE INFO	Display the entered name of the sensor, the sensor serial number, the software version number, and the sensor driver version number.																																		
CAL DATA	Displays the current offset correction and the date of the last calibration.																																		

5.5 Sensor Setup Menu (continued)

SIGNAL	SENSOR SIGNAL allows the user to set the sensor range and display the sensor ADC counts and TEMP ADC COUNTS shows raw data for temperature ADC counts. Comparable to A/D counts.
COUNTERS	SENSOR DAYS displays the cumulative days the sensor has been in use and RESET SENSOR allows the sensor counter to be reset to zero.

5.6 Pressure and Elevation

Note: If the barometric pressure from [Table 6](#) is entered in the meter, the altitude entered in combination with this value must be 0 feet.

[Table 6 on page 25](#) can be used to estimate the true barometric pressure at certain elevations. The correspondence is based on the assumption that at sea level the barometric pressure is 760 mm Hg. After determining the barometric pressure from the table or obtaining it from a local weather service, enter this value into the instrument.

Table 6 Elevation Barometric Pressure

Elevation in feet	Barometric pressure in mm Hg	Elevation in feet	Barometric pressure in mm Hg
0	760	6000	613
500	746	6500	601
1000	733	7000	590
1500	720	7500	579
2000	708	8000	568
2500	695	8500	559
3000	683	9000	548
3500	671	9500	538
4000	659	10000	527
4500	647	10500	517
5000	635	11000	506
5500	624	—	—

5.7 Calibration

Calibration Methods When Measuring Conductivity:

- **Sample Cal Method:** Enter the known conductivity value of the sample determined by laboratory analysis or a comparison reading.
- **Conductivity Cal Method:** Enter the known conductivity value of calibration solution, and its linear % per °C and reference temperature values.
- **Zero Cal Method:** Enter the zero value (in air).

Calibration Methods When Measuring Percent Concentration:

- **Process Cal (Conc) Method:** Enter the known % concentration value of calibration solution (ideally, process solution)
- **Conductivity Cal Method:** Enter the known conductivity value of calibration solution, and its linear % per °C and reference temperature values.
- **Sample Cal Method:** Enter the known % concentration value of the sample determined by laboratory analysis or a comparison reading.

Calibration Methods When Measuring TDS

- **Process Cal (ppm) Method:** Enter the known TDS ppm value of calibration solution (ideally, process solution)
- **Conductivity Cal Method:** Enter the known conductivity value of calibration solution, and its linear % per °C and reference temperature values.
- **Sample Cal Method:** Enter the known TDS ppm value of the sample determined by laboratory analysis or a comparison reading.

5.7.1 Preparing a Conductivity Reference Solution

Add the listed grams of pure, dried NaCl to one liter of high-purity, deionized, CO₂-free water that is at a temperature of 25 °C.

Table 7 Conductivity Reference Solutions

Desired Solution Value			Grams NaCl To Be Added
µS/cm	mS/cm	ppm (NaCl)*	
200	0.20	100	0.1
500	0.5	250	0.25
1000	1.00	500	0.50
2000	2.00	1010	1.01
300	3.00	1530	1.53
4000	4.00	2060	2.06
5000	5.00	2610	2.61
8000	8.00	4340	4.34
10,000	10.00	5560	5.56
20,000	20.00	11,590	11.59
50,000	50.00	31,950	31.95
100,000	100.00	72,710	72.71

5.7.2 Zero Cal

A zero cal will mask interferences when mounting configurations are too close to objects (including the pipe when mounted in-line) in the sample or sample stream.

1. From the Main Menu, select SENSOR SETUP and confirm.
2. Highlight the appropriate sensor if more than one sensor is attached and confirm.
3. Select CALIBRATE and confirm.
4. Select ZERO. Select the available Output Mode (Active, Hold, or Transfer) from the list box and confirm.
5. Move the sensor to air and confirm to continue.
6. The zero calibration procedure will begin and "Wait to Stabilize" will be displayed. The current value and temperature will be displayed. Confirm.
7. Return the sensor to the process.

5.7.3 Sample Cal

1. From the Main Menu, select SENSOR SETUP and confirm.
2. Highlight the appropriate sensor if more than one sensor is attached and confirm.
3. Select CALIBRATE and confirm.
4. Select SAMPLE CAL. Select the available Output Mode (Active, Hold, or Transfer) from the list box and confirm.
5. Move the probe to sample and confirm to continue.
6. Confirm when stable. A Sample Cal complete and the slope will be displayed.
7. Return the sensor to the process.

5.7.4 Cond Cal

1. From the Main Menu, select SENSOR SETUP and confirm.
2. Highlight the appropriate sensor if more than one sensor is attached and confirm.
3. Select CALIBRATE and confirm.
4. Select COND CAL. Select the available Output Mode (Active, Hold, or Transfer) from the list box and confirm.
5. Select SET REF TEMP and confirm.
6. Select SET SLOPE and confirm.
7. Move the probe to the solution and confirm to continue.
8. Confirm when stable. A CAL Complete will be displayed and confirm.
9. Return the sensor to process.

5.7.5 Concurrent Calibration of Two Sensors

1. Begin a calibration on the first sensor and continue until "Wait to Stabilize" is displayed.
2. Select LEAVE and confirm. The display will return to the Main Measurement screen and the reading for both sensors will be flashing.
3. Begin the calibration for the second sensor and continue until "Wait to Stabilize" is displayed.
4. Select LEAVE. The display will return to the Main Measurement screen and the reading for both sensors will be flashing. The calibration for both sensors are now running in the background.
5. To return to the calibration of either sensor select the Main Menu. Select Sensor Setup and confirm. Select the appropriate sensor and confirm.
6. The calibration in progress will be displayed. Continue with the calibration.

Section 6 Maintenance

DANGER

Only qualified personnel should conduct the tasks described in this section of the manual.

DANGER

Seul un technicien qualifié peut effectuer les tâches d'installation décrites dans cette section du manuel.



DANGER

Explosion hazard. Do not connect or disconnect equipment unless power has been switched off or the area is known to be non-hazardous.

DANGER

Risque d'explosion. Couper le courant ou s'assurer que l'emplacement est désigné non dangereux avant de remplacer le ou les composants.

DANGER

Explosion hazard. Substitution of components may impair suitability for Class 1, Division 2.

DANGER

Risque d'explosion. La substitution de composants peut rendre ce matériel inacceptable pour les emplacements de Classe 1, Division 2.

6.1 Maintenance Schedule

Maintenance Task	90 days
Clean the sensor ¹	x
Inspect sensor for damage	x
Calibrate Sensor (if required by regulatory agency)	Per the schedule mandated by your regulatory agency.

¹ Cleaning frequency is application dependent. More or less frequent cleaning will be appropriate in some applications.

6.2 Cleaning the Sensor

Clean the exterior of the sensor with a stream of clean water. If debris remains, use a wet cloth or brush.

7.1 Error Codes

When a sensor is experiencing an error condition, the sensor reading on the measurement screen will flash and all relays and analog outputs associated with this sensor will be held. The following conditions will cause the sensor reading to flash:

- Sensor calibration
- Loss of communication

Highlight the Sensor Diag menu and press **ENTER**. Highlight Errors and press **ENTER** to determine the cause of the error. Errors are defined in [Table 8](#).

Table 8 Error Codes

Displayed Error	Definition	Resolution
ADC FAILURE	A/D conversion failed	Contact Technical and Customer Service (U.S.A. only) on page 35

7.2 Warnings

A Sensor warning will leave all menus, relays, and outputs functioning normally, but will cause a warning icon to flash on the right side of the display. Highlight the Sensor Diag menu and press **ENTER** to determine the cause of the warning.

A warning may be used to trigger a relay and users can set warning levels to define the severity of the warning. Errors are defined in [Table 9](#).

Table 9 Warning Codes

Displayed Warning	Definition	Resolution
PROBE OUT RANGE	No probe connected or measurement reading is out of range.	Verify the solution concentration and ensure that the sensor in use has the correct measurement range.
TEMP OUT RANGE	No temperature element connected or temperature reading is out of range.	Verify that the solution temperature is between –20 to 200 °C.
FLASH FAILURE	Cannot write flash memory.	Contact Technical and Customer Service (U.S.A. only) on page 35.
CAL REQUIRED	Past the calibration days since last cal date.	Calibrate system.
REPLACE SENSOR	Past the total days of current sensor in use. New sensor may need to be installed.	Inspect sensor. If physically damaged or malfunctioning, replace the sensor. Otherwise, reset timer and allow sensor to remain in operation.

Section 8 Replacement Parts and Accessories

8.1 Replacement Items, Accessories, and Reagent and Standards

Item	Quantity	Catalog Number
Cable, sensor extension, 1 m (3.2 ft)	each	61224-00
Cable, sensor extension, 7.7 m (25 ft)	each	57960-00
Cable, sensor extension, 15 m (50 ft)	each	57961-00
Cable, sensor extension, 31 m (100 ft)	each	57962-00
Conductivity Reference Solution, 100–1000 $\mu\text{S}/\text{cm}$	1L	25M3A2000-119
Conductivity Reference Solution, 1000–2000 $\mu\text{S}/\text{cm}$	1L	25M3A2050-119
Conductivity Reference Solution, 2000–150,000 $\mu\text{S}/\text{cm}$	1L	25M3A2100-119
Conductivity Reference Solution, 200,000–300,000 $\mu\text{S}/\text{cm}$	1L	25M3A2200-119
Connector Safety Lock	each	6139900
Digital Gateway for Inductive Conductivity	each	61208-00
Instruction manual, Inductive Conductivity System, English	each	61204-18
Instruction manual, sc100 Controller, English	each	58600-18

Section 9 How to Order

U.S.A. Customers

By Telephone:

6:30 a.m. to 5:00 p.m. MST
Monday through Friday
(800) 227-HACH (800-227-4224)

By Fax:

(970) 669-2932

By Mail:

Hach Company
P.O. Box 389
Loveland, Colorado 80539-0389 U.S.A.

Ordering information by e-mail: orders@hach.com

Information Required

- Hach account number (if available)
- Your name and phone number
- Purchase order number
- Brief description or model number
- Billing address
- Shipping address
- Catalog number
- Quantity

International Customers

Hach maintains a worldwide network of dealers and distributors. To locate the representative nearest you, send an e-mail to: intl@hach.com or contact:

Hach Company World Headquarters; Loveland, Colorado, U.S.A.

Telephone: (970) 669-3050; Fax: (970) 669-2932

Technical and Customer Service (U.S.A. only)

Hach Technical and Customer Service Department personnel are eager to answer questions about our products and their use. Specialists in analytical methods, they are happy to put their talents to work for you.

Call 1-800-227-4224 or e-mail techhelp@hach.com

Section 10 Repair Service

Authorization must be obtained from Hach Company before sending any items for repair. Please contact the Hach Service Center serving your location.

In the United States:

Hach Company
Ames Service
100 Dayton Avenue
Ames, Iowa 50010
(800) 227-4224 (U.S.A. only)
FAX: (515) 232-3835

In Canada:

Hach Sales & Service Canada Ltd.
1313 Border Street, Unit 34
Winnipeg, Manitoba
R3H 0X4
(800) 665-7635 (Canada only)
Telephone: (204) 632-5598
FAX: (204) 694-5134
E-mail: canada@hach.com

**In Latin America, the Caribbean, the Far East,
Indian Subcontinent, Africa, Europe, or the Middle East:**

Hach Company World Headquarters,
P.O. Box 389
Loveland, Colorado, 80539-0389 U.S.A.
Telephone: (970) 669-3050
FAX: (970) 669-2932
E-mail: intl@hach.com

Section 11 Limited Warranty

Hach Company warrants its products to the original purchaser against any defects that are due to faulty material or workmanship for a period of one year from date of shipment unless otherwise noted in the product manual.

In the event that a defect is discovered during the warranty period, Hach Company agrees that, at its option, it will repair or replace the defective product or refund the purchase price excluding original shipping and handling charges. Any product repaired or replaced under this warranty will be warranted only for the remainder of the original product warranty period.

This warranty does not apply to consumable products such as chemical reagents; or consumable components of a product, such as, but not limited to, lamps and tubing.

Contact Hach Company or your distributor to initiate warranty support. Products may not be returned without authorization from Hach Company.

Limitations

This warranty does not cover:

- Damage caused by acts of God, natural disaster, labor unrest, acts of war (declared or undeclared), terrorism, civil strife or acts of any governmental jurisdiction
- Damage caused by misuse, neglect, accident or improper application or installation
- Damage caused by any repair or attempted repair not authorized by Hach Company
- Any product not used in accordance with the instructions furnished by Hach Company
- Freight charges to return merchandise to Hach Company
- Freight charges on expedited or express shipment of warranted parts or product
- Travel fees associated with on-site warranty repair

This warranty contains the sole express warranty made by Hach Company in connection with its products. All implied warranties, including without limitation, the warranties of merchantability and fitness for a particular purpose, are expressly disclaimed.

Some states within the United States do not allow the disclaimer of implied warranties and if this is true in your state the above limitation may not apply to you. This warranty gives you specific rights, and you may also have other rights that vary from state to state.

This warranty constitutes the final, complete, and exclusive statement of warranty terms and no person is authorized to make any other warranties or representations on behalf of Hach Company.

Limitation of Remedies

The remedies of repair, replacement or refund of purchase price as stated above are the exclusive remedies for the breach of this warranty. On the basis of strict liability or under any other legal theory, in no event shall Hach Company be liable for any incidental or consequential damages of any kind for breach of warranty or negligence.

Section 12 Compliance Information

Hach Co. certifies this instrument was tested thoroughly, inspected and found to meet its published specifications when it was shipped from the factory.

The **Model sc100 Controller/sc1000 Controller with the Inductive Conductivity Sensor** has been tested and is certified as indicated to the following instrumentation standards:

Product Safety

UL 61010A-1 (ETL Listing # 65454)
CSA C22.2 No. 1010.1 (ETLc Certification # 65454)
Certified by Hach Co. to EN 61010-1 Amds. 1 & 2 (IEC1010-1) per 73/23/EEC, supporting test records by Intertek Testing Services.

Immunity

This equipment was tested for industrial level EMC per:

EN 61326 (EMC Requirements for Electrical Equipment for Measurement, Control and Laboratory Use) **per 89/336/EEC EMC:** Supporting test records by Hach Company, certified compliance by Hach Company.

Standards include:

IEC 1000-4-2:1995 (EN 61000-4-2:1995) Electrostatic Discharge Immunity (Criteria B)
IEC 1000-4-3:1995 (EN 61000-4-3:1996) Radiated RF Electromagnetic Field Immunity (Criteria A)
IEC 1000-4-4:1995 (EN 61000-4-4:1995) Electrical Fast Transients/Burst (Criteria B)
IEC 1000-4-5:1995 (EN 61000-4-5:1995) Surge (Criteria B)
IEC 1000-4-6:1996 (EN 61000-4-6:1996) Conducted Disturbances Induced by RF Fields (Criteria A)
IEC 1000-4-11:1994 (EN 61000-4-11:1994) Voltage Dip/Short Interruptions (Criteria B)

Additional Immunity Standard/s include:

ENV 50204:1996 Radiated Electromagnetic Field from Digital Telephones (Criteria A)

Emissions

This equipment was tested for Radio Frequency Emissions as follows:

Per **89/336/EEC EMC: EN 61326:1998** (Electrical Equipment for measurement, control and laboratory use—EMC requirements) Class “A” emission limits. Supporting test records by Hewlett Packard, Fort Collins, Colorado Hardware Test Center (A2LA # 0905-01) and certified compliance by Hach Company.

Standards include:

EN 61000-3-2 Harmonic Disturbances Caused by Electrical Equipment
EN 61000-3-3 Voltage Fluctuation (Flicker) Disturbances Caused by Electrical Equipment

Additional Emissions Standard/s include:

EN 55011 (CISPR 11), Class “A” emission limits

Canadian Interference-causing Equipment Regulation, IECS-003, Class A

Supporting test records by Hewlett Packard, Fort Collins, Colorado Hardware Test Center (A2LA # 0905-01) and certified compliance by Hach Company.

This Class A digital apparatus meets all requirements of the Canadian Interference-Causing Equipment Regulations.

Cet appareil numérique de la classe A respecte toutes les exigences du Règlement sur le matériel brouilleur du Canada.

FCC PART 15, Class “A” Limits

Supporting test records by Hewlett Packard, Fort Collins, Colorado Hardware Test Center (A2LA # 0905-01) and certified compliance by Hach Company.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Changes or modifications to this unit not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at his own expense. The following techniques of reducing the interference problems are applied easily.

1. Disconnect the Controller from its power source to verify that it is or is not the source of the interference.
2. If the Controller is connected into the same outlet as the device with which it is interfering, try another outlet.
3. Move the Controller away from the device receiving the interference.
4. Reposition the receiving antenna for the device receiving the interference.
5. Try combinations of the above.

Appendix A Modbus Register Information

Table 10 Sensor Modbus Registers

Group Name	Tag Name	Register #	Data Type	Length	R/W	Description
Tags	SensorMeasTag	40001	Integer	1	R	Sensor Measurement Tag
Measurements	DOMeas	40002	Float	2	R	DO Measurement
Tags	TempMeasTag	40004	Integer	1	R	Temperature Measurement Tag
Measurements	TempDegCMeas	40005	Float	2	R	Temperature Measurement
Configuration	SensorName	40007	String	6	R/W	Sensor Name
Tags	FuncCode	40013	Integer	1	R/W	Function Code tag
Tags	NextState	40014	Integer	1	R/W	Next State Tag
Configuration	TempUnits	40015	Integer	1	R/W	Temperature Units—C or F
Configuration	Filter	40016	Integer	1	R/W	Sensor Filter
Configuration	TempElementType	40017	Integer	1	R/W	Temperature Element Type
Tags	TempUserValueTag	40018	Integer	1	R	Temperature User Value Tag
Configuration	TempUserDegCValue	40019	Float	2	R/W	Temperature User Value
Configuration	PressureUnits	40021	Integer	1	R/W	Pressure Units
Configuration	SalinityUnits	40022	Integer	1	R/W	Salinity Units
Tags	PressureTag	40023	Integer	1	R	Pressure Tag
Configuration	Pressure	40024	Float	2	R/W	Pressure
Tags	SalinityTag	40026	Integer	1	R	Salinity Units
Configuration	Salinity	40027	Float	2	R/W	Salinity
Configuration	MeasUnits	40029	Integer	1	R/W	Measurement Units
Calibration	OutputMode	40030	Integer	1	R/W	Output Mode
Calibration	CalLeave	40031	Integer	1	R/W	Cal Leave Mode
Calibration	CalAbort	40032	Integer	1	R/W	Cal Abort Mode
Tags	CalEditValueTag	40033	Integer	1	R	Cal Edit Value Tag
Calibration	CalEditDOValue	40034	Float	2	R/W	Cal Edit Value
Diagnostics	SoftwareVersion	40036	String	6	R	Software Version
Diagnostics	SerialNumber	40042	String	6	R	Serial Number
Diagnostics	CalQValue	40048	Float	2	R	DO Cal Q Value
Calibration	CalCode	40050	Integer	1	R	Cal Code
Configuration	SensorLogInterval	40051	Integer	1	R/W	Sensor Data Log Interval
Configuration	TempLogInterval	40052	Integer	1	R/W	Temperature Data Log Interval
Diagnostics	DOmV	40053	Float	2	R	DO mV
Diagnostics	ProdDate	40055	Date	2	R/W	Production Date
Diagnostics	LastCalDate	40057	Date	2	R	Last Calibration Date
Diagnostics	SensorDays	40059	Integer	1	R	Sensor Running Days
Configuration	RejectFrequency	40060	Integer	1	R/W	Reject Frequency
Diagnostics	DeviceDriver	40061	String	5	R	Device Driver
Configuration	CalWarningDays	40066	Integer	1	R/W	Calibration Warning Days
Configuration	SensorWarningDays	40067	Integer	1	R/W	Sensor Warning Days
Tags	SensorMeasTag	40001	Integer	1	R	Sensor Measurement Tag
Measurements	DOMeas	40002	Float	2	R	DO Measurement
Tags	TempMeasTag	40004	Integer	1	R	Temperature Measurement Tag

Table 10 Sensor Modbus Registers (continued)

Group Name	Tag Name	Register #	Data Type	Length	R/W	Description
Measurements	TempDegCMeas	40005	Float	2	R	Temperature Measurement
Configuration	SensorName	40007	String	6	R/W	Sensor Name
Tags	FuncCode	40013	Integer	1	R/W	Function Code tag
Tags	NextState	40014	Integer	1	R/W	Next State Tag
Configuration	TempUnits	40015	Integer	1	R/W	Temperature Units—C or F
Configuration	Filter	40016	Integer	1	R/W	Sensor Filter
Configuration	TempElementType	40017	Integer	1	R/W	Temperature Element Type
Tags	TempUserValueTag	40018	Integer	1	R	Temperature User Value Tag
Configuration	TempUserDegCValue	40019	Float	2	R/W	Temperature User Value
Configuration	PressureUnits	40021	Integer	1	R/W	Pressure Units
Configuration	SalinityUnits	40022	Integer	1	R/W	Salinity Units
Tags	PressureTag	40023	Integer	1	R	Pressure Tag
Configuration	Pressure	40024	Float	2	R/W	Pressure
Tags	SalinityTag	40026	Integer	1	R	Salinity Units
Configuration	Salinity	40027	Float	2	R/W	Salinity
Configuration	MeasUnits	40029	Integer	1	R/W	Measurement Units
Calibration	OutputMode	40030	Integer	1	R/W	Output Mode
Calibration	CalLeave	40031	Integer	1	R/W	Cal Leave Mode
Calibration	CalAbort	40032	Integer	1	R/W	Cal Abort Mode
Tags	CalEditValueTag	40033	Integer	1	R	Cal Edit Value Tag
Calibration	CalEditDOValue	40034	Float	2	R/W	Cal Edit Value
Diagnostics	SoftwareVersion	40036	String	6	R	Software Version
Diagnostics	SerialNumber	40042	String	6	R	Serial Number
Diagnostics	CalQValue	40048	Float	2	R	DO Cal Q Value
Calibration	CalCode	40050	Integer	1	R	Cal Code
Configuration	SensorLogInterval	40051	Integer	1	R/W	Sensor Data Log Interval
Configuration	TempLogInterval	40052	Integer	1	R/W	Temperature Data Log Interval
Diagnostics	DOmV	40053	Float	2	R	DO mV
Diagnostics	ProdDate	40055	Date	2	R/W	Production Date
Diagnostics	LastCalDate	40057	Date	2	R	Last Calibration Date
Diagnostics	SensorDays	40059	Integer	1	R	Sensor Run\ning Days
Configuration	RejectFrequency	40060	Integer	1	R/W	Reject Frequency
Diagnostics	DeviceDriver	40061	String	5	R	Device Driver
Configuration	CalWarningDays	40066	Integer	1	R/W	Calibration Warning Days
Configuration	SensorWarningDays	40067	Integer	1	R/W	Sensor Warning Days
Tags	SensorMeasTag	40001	Integer	1	R	Sensor Measurement Tag
Measurements	DOMeas	40002	Float	2	R	DO Measurement

B

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